Dr. Deviprasad Sahoo

Postdoctoral Researcher, Experimental Physicist
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PROFILE

Experimental physicist (Ph.D. in Physics, 3+ years of postdoctoral experience) specialising in ultrafast optics, light–matter interaction, materials science and energy-harvesting devices. I build custom ultrafast optical systems, and synthesise semiconductor thin films and nanomaterials, for next-generation photonics and energy applications. I also fabricate devices and run multiphysics simulations. Output to date: 28 peer-reviewed publications (20 first-author), ~1000 citations, h-index 18, and MSCA CZ awardee.

RESEARCH EXPERIENCE

Postdoctoral Researcher — Attosecond Photonics **Mar 2026 – present**
HUN-REN Wigner Research Centre for Physics, Budapest, Hungary

- Built a TIPTOE field-sampling setup for attosecond metrology, driven by a ~ Ti: sapphire oscillator
- Performed attosecond spectroscopy by using tunnelling-ionization mechanisms in III–V semiconductors.
- Optimised time- and field-resolved nonlinear-optical metrology: Z-scan, chirp correction and CEP-dependent measurements for petahertz photonics.

Postdoctoral Fellow — Femtosecond Photonics **Oct 2025 – Mar 2026**
Indian Institute of Technology Madras, Department of Physics, India

- Developed a custom pump – probe set-up for transient-absorption spectroscopy for 2D MXene characterization.
- Performed high-harmonic generation experiments in an ultra-high-vacuum beamline driven by a 30 fs CPA Ti:sapphire oscillator.

Postdoctoral Research Associate — Energy Materials & Devices **Dec 2023 – Oct 2025**
CIPET: SARP – LARPM (R&D), Bhubaneswar, India

- Fabricated wearable PVDF–MXene nanofibre self-powered sensors (TENG/PENG); benchmarked 2D nanofillers (MXene, hBN, rGO), published in *Advanced Materials Technologies* (2025).
- Synthesised and characterised MXene 2D materials (Ti_3C_2, Nb_2C, Ti_2N, Ti_3CN) via etching routes (HF, LiF, hydrothermal) for HER, OER and photodetection.
- Built a YOLO + ByteTrack computer-vision pipeline (Python, Ultralytics, PyTorch, OpenCV) with pixel-to-mm calibration for automated bubble detection.
- Synthesised 3 polymer-composite systems (PDMS, PEDOT:PSS, cellulose aerogel) for TENG/PENG energy harvesting; simulated and prototyped devices in COMSOL Multiphysics and CAD.

Postdoctoral Research Associate **Jul 2023 – Dec 2023**
Fakir Mohan University, Balasore, India

- Characterised energy-storage materials by LSV, CV and electrochemical impedance spectroscopy.

EDUCATION

Ph.D. in Physics **2019 – 2023**
Institute of Chemical Technology, Mumbai, India | Supervisor: Dr. Ramakanta Naik
Thesis: *Structural, morphological and optical study of doped As–Se films under different energy treatments*

- Fabricated As–Se chalcogenide semiconductor thin films by thermal and e-beam evaporation; characterised them across structural, morphological and nonlinear-optical domains.
- Engineered defects through controlled energy–matter routes: laser irradiation, low-energy ion implantation and thermal annealing.
- Performed pump–probe and fluence-dependent ion-irradiation experiments; measured nonlinear optical parameters and designed optical setups.
- Studied frequency- and temperature-dependent dielectric and impedance response of hydrothermally synthesised Cu–Ni–Sn–S chalcopyrite materials.

M.Sc. in Physics: *Fakir Mohan University, Balasore, India* **2016 – 2018**

B.Sc. (Hons.) in Physics: Maharishi College of Natural Law, Utkal University, Bhubaneswar, India **2013 – 2016**

TECHNICAL SKILLS

- **Ultrafast optics:** few-cycle CEP-stable Ti:sapphire oscillators (sub-5.5 fs, 800 nm, 80 MHz); CPA amplifiers (800 nm, 30 fs); TIPTOE field sampling; high-harmonic generation in gases; transient-absorption spectroscopy; pump–probe and autocorrelation; Z-scan; femtosecond beam metrology and alignment.

- **Materials synthesis:** thermal and e-beam evaporation, electrospinning, hydro/solvothermal synthesis, MXene etching (HF, LiF, hydrothermal).
- **Characterisation:** XRD, Raman, UV-Vis-NIR, FTIR, FESEM, HR-TEM, TGA, electrochemical workstation (CV, LSV, EIS), low-energy ion implantation.
- **Device fabrication:** triboelectric and piezoelectric nanogenerators, wearable pulse sensors, photodetectors, supercapacitors, FDM 3D printing.
- **Signal detection & electronics:** transimpedance amplification, low-noise preamplification, source-meter, electrometer, LCR meter.
- **Modelling & code:** Python, MATLAB, machine learning and computer vision; COMSOL Multiphysics, Ansys, Fusion 360, SolidWorks; C/C++.
- **Laboratory build-out:** specifying, procuring and commissioning optical and electronic instruments.

AWARDS / FELLOWSHIPS / ADDITIONAL INFORMATION

- Marie Curie Postdoctoral Fellowships (MSCA CZ) – 2025, European Commission.
- International Travel Support (ITS) grant — ANRF/SERB, Government of India, for ICEAN 2024, University of Newcastle, Australia.
- Young Achiever Award presentation — DAE Solid State Physics Symposium 2024, BARC, India.
- Peer reviewer (2 manuscripts refereed) — *Journal of Non-Crystalline Solids*, *Materials Letters*.
- Guest Topical Editor — *The Open Biomedical Engineering Journal*.
- Mentored 3 students.
- Affiliate Member — Royal Society of Chemistry (743537).

PUBLICATIONS

28 peer-reviewed journal articles (15 as first author | 4 reviews | 4 book chapters | ~1000 citations, h-index 18, i10-index 21 | cumulative impact factor >100.

Selected Journal Articles

1. **Sahoo D**, Sahoo S, Nayak D, Mohanty S, Naik R. Enhancing triboelectric performance through 2D nanofillers (MXene, hBN, rGO) in PVDF nanofibre for self-powered sensing. *Advanced Materials Technologies*, 2025, e01141.
2. Nayak D et al. (incl. **Sahoo D**). ZnO-enhanced FRET-sensitised PMMA nanocomposites for efficient photocatalysis and antibacterial activity. *Surfaces and Interfaces*, 2025, 64, 106354.
3. **Sahoo D** et al. Phase-change-driven tuning of optical properties of Te/As_2Se_3 bilayer thin films. *Materials Science & Engineering B*, 2023, 297, 116750.
4. **Sahoo D** et al. Hydrothermally synthesised $Cu_{0.06-x}Ni_{0.03}Sn_{0.03+x}S_{0.12}$ nanosheets for optoelectronic and dielectric applications. *Materials Today Electronics*, 2023, 3, 100030.
5. **Sahoo D** et al. Optical and dielectric characterisation of $Cu_2Ni_{1+x}Sn_{1-x}S_4$ nanospheres. *ACS Applied Engineering Materials*, 2023, 1, 1001.
6. **Sahoo D**, Naik R. Progress and prospects of 2D VS_2 transition-metal dichalcogenides. *FlatChem*, 2022, 36, 100455. (review)
7. **Sahoo D**, Naik R. Review on the optical properties of Se-doped chalcogenide thin films. *Journal of Non-Crystalline Solids*, 2022, 597, 121934. (review)
8. **Sahoo D**, Naik R. GSST phase-change materials and their utilisation in optoelectronic devices: a review. *Materials Research Bulletin*, 2022, 148, 111679. (review)
9. **Sahoo D**, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Impact of irradiation doses on the structural, morphological and linear–nonlinear optical properties of $Ge_{10}Sb_{25}Se_{65}$ thin films for optoelectronic applications. *European Physical Journal Plus*, 2022, 137, 707.
10. **Sahoo D**, Sahoo S, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Proton ion irradiation on $As_{40}Se_{50}Sb_{10}$ thin films: fluence-dependent tuning of linear–nonlinear optical properties for photonic applications. *ACS Applied Electronic Materials*, 2022, 4, 856–868.
11. **Sahoo D** et al. *Indian Journal of Physics*, 2022.
12. **Sahoo D**, Priyadarshini P, Dandela R, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. In-situ laser irradiation: the kinetics of the changes in the nonlinear/linear optical parameters of $As_{50}Se_{40}Sb_{10}$ thin films for photonic applications. *RSC Advances*, 2021, 11, 16015–16025.
13. **Sahoo D**, Priyadarshini P, Aparimita A, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Optimisation of linear and nonlinear optical parameters of $As_{40}Se_{50}Te_{10}$ thin films by thermal annealing. *Optics & Laser Technology*, 2021, 140, 107036.
14. **Sahoo D** et al. Investigation of amorphous–crystalline transformation induced optical and electronic properties change in annealed $As_{50}Se_{50}$ thin films. *Optical and Quantum Electronics*, 2021, 53, 197.
15. **Sahoo D**, Priyadarshini P, Dandela R, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Optimisation of linear and nonlinear optical parameters in $As_{40}Se_{60}$ film by annealing at different temperatures. *Optik*, 2020, 219, 165286.
16. **Sahoo D**, Priyadarshini P, Aparimita A, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Role of annealing temperature on optimising the linear and nonlinear optical properties of $As_{40}Se_{50}Ge_{10}$ films. *RSC Advances*, 2020, 10, 26675–26685.

17. Parida A, **Sahoo D**, Alagarasan D, Vardhrajaperumal S, Ganesan R, Naik R. Enhancing the third-order nonlinearity and crystallinity by selenium incorporation in tin sulfide films ($SnS_{1-x}Se_x$) for optoelectronic applications. *Materials Advances*, 2022, 3, 5930–5940.
18. Parida A, **Sahoo D**, Alagarasan D, Vardhrajaperumal S, Ganesan R, Naik R. Increase in nonlinear susceptibility and refractive index in quaternary $In_{15}Sb_{10}S_{15}Se_{60}$ thin films upon annealing at different temperatures for photonic applications. *Journal of Alloys and Compounds*, 2022, 905, 164143.
19. Parida A, **Sahoo D**, Alagarasan D, Varadharajaperumal S, Ganesan R, Naik R. Impact on nonlinear/linear optical and structural parameters in quaternary $In_{15}Ag_{10}S_{15}Se_{60}$ thin films upon annealing at different temperatures. *Ceramics International*, 2022, 48, 15380–15389.
20. Priyadarshini P, **Sahoo D**, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Structural and optoelectronic properties change in Bi/In_2Se_3 heterostructure films by thermal annealing and laser irradiation. *Journal of Applied Physics*, 2021, 129, 223101.
21. Priyadarshini P, **Sahoo D**, Naik R. A review on the optical properties of some germanium-based chalcogenide thin films and their applications. *Optical and Quantum Electronics*, 2022, 54, 166. (review)
22. Priyadarshini P, **Sahoo D**, Aparimita A, Alagarasan D, Ganesan R, Varadharajaperumal S, Naik R. Switching of linear and nonlinear optical parameters in $As_{35}Se_{65}$ thin films upon annealing at both above and below T_g. *Applied Physics A*, 2020, 126, 910.

**Two manuscripts currently under review.*

Conference Proceedings

1. **Sahoo D** et al. *AIP Conference Proceedings*, 2020, 2265(1), 030236.
2. **Sahoo D** et al. *Materials Today: Proceedings*, 2023.

Book Chapters

1. **Sahoo D**, Naik R. In: *Handbook of Energy Materials*. Springer, 2022.
2. **Sahoo D**, Naik R. In: *Smart Nanomaterials*. Cambridge Scholars, accepted.
3. **Sahoo D** et al. In: *Theory and Applications of Nonparenteral Nanomedicines*. Academic Press, 2021.
4. Bandaru R et al. (incl. **Sahoo D**). In: *Nanotechnology for Tuberculosis Treatment*. Academic Press, 2020.

CONFERENCE PRESENTATIONS

- Extreme Light Infrastructure (ELI), Prague, Czech Republic — presentation.
- ICEAN 2024, University of Newcastle, Australia — invited international presentation (ANRF/SERB travel grant).
- DAE Solid State Physics Symposium — BARC Mumbai 2024 (Young Achiever Award presentation), BARC Mumbai 2021, IIT Jodhpur 2019.
- APA–EPNOE–GFL International Conference, Jaipur, India — invited presentation.
- ISSMD 2022, KIIT Bhubaneswar; ORC 2023, Sambalpur University; Science and Technology of Functional Materials 2019, SOA University, Odisha.

REFEREES

Refrees details will be available upon request.